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Topic 2 Part 1. Introduction to R

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Introduction to (one lecture)

- ✓ Using the **R** programming language via the **RStudio** interface
- ✓ Hands-on DataCamp training and Computer lab classes will give you the **practical skills** to carry out statistical analyses using R

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What is

- R is a programming language for statistical computing and visualisation
- R software is open-source and free
- Commonly used software for statistics
- ~25,000 contributed packages / libraries
- Lots of tutorials online and online help
- Help in labs

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2.1 What are R and RStudio?

moderndive.com

For much of this book, we will assume that you are using R via RStudio. First time users often confuse the two. At its simplest:

- R is like a car's engine
- RStudio is like a car's dashboard

R: Engine



RStudio: Dashboard



More precisely, R is a programming language that runs computations while RStudio is an *integrated development environment (IDE)* that provides an interface by adding many convenient features and tools. So the way of having access to a speedometer, rearview mirrors, and a navigation system makes driving much easier, using RStudio's interface makes using R much easier as well.

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Getting started with R and RStudio

On your laptop

1. Download and install R (the “engine”)
2. Download and install RStudio (the “dashboard”)

On college PCs

1. Locate the R app and install
2. Locate the RStudio app and install

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<https://cran.r-project.org>

+



<https://posit.co/downloads>

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Now you have two new programs (apps) on your laptop/PC

R: Do not open this



RStudio: Open this



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 **Studio** is an integrated development environment for R.

Commands stored in an R script file.

R session window: Processed commands and outputs.

Details of Data Objects produced.

Graphics produced

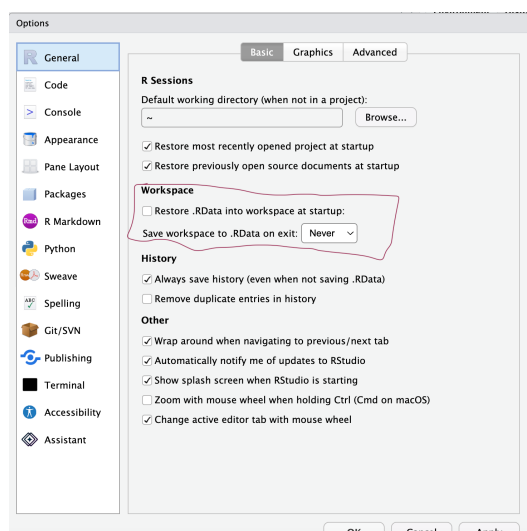
Send commands to R session window using Run or Source

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RStudio options setup

- With RStudio open
- Go to **Tools > Global Options**, clear the tick mark from “Restore .RData into Workspace at Startup” and choose **Never** on the “Save workspace to .RData on exit”.



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RStudio workflow

- We'll create an R Project for ST2001 labs
- Make a new R project: **File>New Project**
- In the popup select **New directory > New Project**
- Give the new project a name (eg ST2001_labs) in the dialogue box, and browse the directories to save the new project in a **folder where you will keep all your ST2001 work**
- **Make sure you create your ST2001 folder in a location you can find again!**
- For example, on OneDrive, in your laptop documents, in your personal file storage on college system
- After you close your R project you can return to it simply by navigating to your ST2001 folder and clicking on the R project app

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How do I code in R?

- R is a statistical programming language
- Not point-and-click
- You do not need to be a programmer to code in R!
- You do need some basic programming concepts
- The DataCamp “Introduction to R” course will help you learn basic programming concepts

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Basic concepts

Term	Meaning
console	Where we enter commands in R
run code	Tell R to carry out some task by entering commands in the console
objects	Where values are saved in R
data types	numeric, logical (TRUE/FALSE), character (in “ ”)
vectors	A series of individual values. Create a vector using the function c for combine, separate the values by commas, eg <code>>c(1,2,3,4,5)</code> creates a vector containing these five numeric values
factors	Categorical data are often stored as factors
data frames	Rectangular array of data with a row for each observation and a column for each variable

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Basic concepts

Term	Meaning
Assignment <- or =	myvector<-c(1,2,3,4,5) stores the values in a vector object called myvector myvector = c(1,2,3,4,5) does the same
Viewing values in an object	Typing myvector in the console will return >myvector [1] 1 2 3 4 5
Function (or command)	A function does a specific task in R. A typical function takes some input(s) and returns output(s), eg the mean() function takes an object and returns the mean >mean(myvector) [1] 3
Boolean operators and TRUE/FALSE	== is equal to, eg 2+3==6 will return FALSE != is not equal to, 2+3!=6 will return TRUE < less than, <= less than or equal to, >, >= ...
Logicals	& and or, eg (2>1) & (1+1==2) will return TRUE

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Errors, warnings and messages

Errors in **red**: something is wrong. For example, you type in the console:

```
>2+2=5
```

```
Error in 2 + 2 = 5 : target of assignment expands to  
non-language object
```

Don't panic – try to figure out what's going on and fix it - maybe you meant to type:

```
>2+2 ==5
```

```
[1] FALSE
```

- Warnings in **yellow** – code will still run but double check if the warning matters
- Messages in **green** – these are just for information

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Tips on learning to code in R

- It's like learning a foreign language – practise to become more familiar, and don't worry about making mistakes, everyone makes mistakes – learn from them
- Computers are not clever! You need to give them very precise instructions on exactly what to do in exactly the way they need to be told – they are very fussy about being given precise instructions
- Copy, paste and modify code to do relevant tasks rather than writing from scratch. This is easier and faster. When you become more confident and familiar you can try writing code from scratch.
- Learn by analysing a data set of interest to you
- Practise, practise, practise

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What are R packages?

- Base R has a large number of **functions** that are available when you start R – eg the `mean()` function that calculates the mean
- Packages (also called libraries) extend the functionality of R
- Packages usually include a **set of functions, data sets, and help files geared towards carrying out specific task** or group of related tasks
- Base R is like a new mobile phone, and packages are like the additional apps you can download to your mobile phone
- You need to download (or install) the package first
- Then you need to open (or load) the package in your R session

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R package

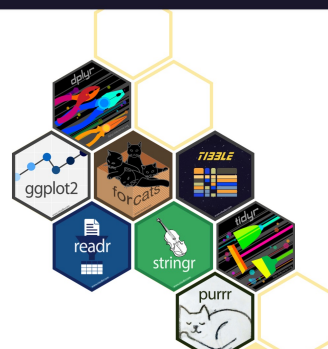
Article [Talk](#)

From Wikipedia, the free encyclopedia

R packages are [extensions](#) to the [R statistical programming language](#). R packages contain code, data, and documentation in a standardised collection format that can be installed by users of R, typically via a centralised [software repository](#) such as CRAN (the Comprehensive R Archive Network).^{[1][2]} The large number of packages available for R, and the ease of installing and using them, has been cited as a major factor driving the widespread adoption of the language in [data science](#).^{[3][4][5][6]}

Tidyverse

[Packages](#) [Blog](#) [Learn](#) [Help](#) [Contribute](#)



R packages for data science

The tidyverse is an opinionated [collection of R packages](#) designed for data science. All packages share an underlying design philosophy, grammar, and data structures.

Install the complete tidyverse with:

```
install.packages("tidyverse")
```

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Installing and loading packages

You only need to **install** a package ONCE

Eg, to install the `ggplot2` package (for nice plots) type in the console:

```
>install.packages("ggplot2")
```

Now, **every time you start a new R session, you will need to load** the packages you want that you have already installed.

Load the `ggplot2` package by running the following:

```
>library(ggplot2)
```

You can then use all the plotting functions in `ggplot2` in your current R session

If you forget to load the package and try to run a function from `ggplot2` you will get an error message.

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A Gentle Start in

Introduction to R

- R as a Calculator
- Storing Things in R
- Vectors to Store Data
- Selecting Data from Objects
- What Have I Created? How to Delete Things?
- Something Fun

R Code

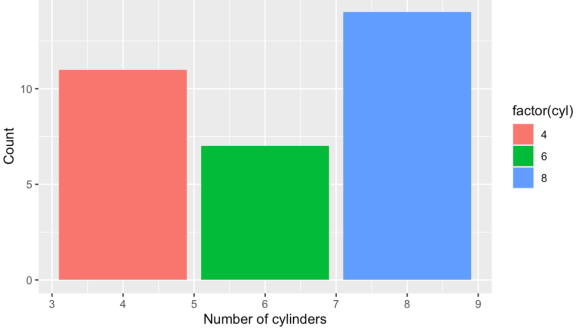
Start Over

Solution

Run Code

```
1 library(tidyverse)
2
3 mtcars %>% ggplot(aes(cyl, fill = factor(cyl))) + geom_bar() +
4   labs(x = "Number of cylinders", y = "Count", title = "Count Cars with No. of Cylinders")
```

Count Cars with No. of Cylinders



Number of cylinders	Count
4	11
6	7
8	14

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The impact of data science and analytics

Data science and analytics are rapidly shaping every aspect of our lives and our businesses. We're collecting more data than ever before, but not everyone is able to efficiently analyze all that data to extract meaningful insights. There is incredible power in data—but only if you know what to do with it. DataCamp teaches companies and individuals the skills they need to work with data in the real world.

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Introduction to R

Basic

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Topic 2 (1, intro to R

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Datacamp statement of accomplishment



Sign up link on
canvas homepage
& in Welcome
Announcement

[DataCamp ST2001 classroom](#)

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Installing  and 

Tutorials on installing R and RStudio on your computer

<https://rstudio-education.github.io/hopr/starting.html>

<https://www.datacamp.com/tutorial/r-studio-tutorial>

More instructions videos on Canvas, but do also google!

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Summary of tasks

- Download and install `R` and `RStudio` on your laptop and college PC profile
- Set the `RStudio` options so you start each `R` session with a blank slate (slide 9)
- Make a **folder for ST2001 labs** on your laptop and college PC profile and/or OneDrive
- Make an R Project for S2001 labs
- Install and load the `tidyverse` package
- We'll go over all these in the first lab class

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